

GE-Problem Set 3, ECN 714

Exercise 1 Consider a model with a single firm, a single consumer, and L goods. (This could be a representative agent model.) Assume that the consumption set, X is the positive orthant, and that the consumer's utility function is continuous, strongly monotone, and quasiconcave, and that her initial endowment $\omega \gg 0$. Assume that the firm's production set Y is closed and convex, and that $0 \in Y$ (i.e. it is feasible to stop producing).

Finally, assume that the production possibility set $\hat{Y} = (\omega + Y) \cap \mathbb{R}_+^L$ is bounded. The production possibility set is the set of output plans that can be produced with the available endowments in the economy.

- Prove directly that the Second Welfare Theorem holds, i.e. prove that every Pareto Optimal allocation in this economy can be supported as a Walrasian equilibrium with transfers.
- Prove directly that a competitive equilibrium exists in this model.

Prove directly means do not just invoke a theorem.

Exercise 2 Let $X = \mathbb{R}$, $Y = \mathbb{R}^2$, and define the correspondence $\varphi : X \rightarrow Y$ by

$$\varphi(x) = \left\{ (y_1, y_2) \in Y : y_2 \geq 0, (\forall y_1 \neq x) \left[y_2 \geq \frac{1}{(y_1 - x)} \right] \right\}.$$

- Graph $\varphi(1)$.
- Is φ closed valued?
- Is φ closed?
- Is φ uhc at the point $x = 1$? Is φ uhc?
- Suppose $Y = [-1, 10]$. Is φ uhc when evaluated at $x = 1$? Is φ uhc?

Note: Most of the solution follows from the graph in (a)

Exercise 3 Consider the preference relation on $\mathbb{R}_+^3$ represented by the utility function

$U(x_1, x_2, x_3) = \sqrt{x_1} + \sqrt{x_2} + x_2 + \frac{x_3}{1+x_3}$, and let the consumer initial endowment be $\omega = (1, 1, 1)$.

- Show that U is strongly monotone, strictly concave, and continuous.
- If $(x_1, x_2, x_3) \in \mathbb{R}_+^3$ and $x_3 > 0$, show that $U(x_1, x_2 + x_3, 0) > U(x_1, x_2, x_3)$.
- If $p = (p_1, p_2, p_3) \gg 0$ and $p_2 = p_3$, show that $x_3(p) = 0$.
- For each n , let $p^n = (1 - \frac{2}{n}, \frac{1}{n}, \frac{1}{n})$. Show that $x_3(p^n) = 0$ for each n (and thus that demand for x_3 remains bounded even though $p_3^n \rightarrow 0$).

e. Show that $\lim_{n \rightarrow \infty} x_2(p^n) = \infty$.

f. Evaluate the following statement. *The price of a good may tend to zero but the demand for "that" good may remain bounded.* If the statement were true, would the boundary condition be violated? Provide a brief explanation.

Exercise 4 Consider a two-good economy. Draw four functions $z : \Delta^\circ \rightarrow \mathbb{R}^2$ demonstrating that if any one of the four conditions stated below fails, then there may not be a solution to $z(p) = 0$. The conditions are (1) continuity, (2) Walras' Law, (3) boundedness below, (4) the boundary condition. Each function you draw should violate only one of the four conditions, and have the property that $(\nexists p) \quad z(p) = 0$.